

SV16 Rev A FPGA board - routing report (DRAFT, not a fabrication release)

Source placement: *sv16_board_placed_native.kicad_pcb* (placement only, 0 copper). Routed output: *sv16_board_routed_draft.kicad_pcb*. Stackup: F.Cu signals, In1.Cu GND plane, In2.Cu 3V3 plane, B.Cu signals (4 layers, as specified in the handoff).

Status: routing is NOT complete. 107 of 119 nets are fully connected; 22 pad connections are open (see section 4). The board must not be sent for fabrication. KiCad DRC, the schematic cross-check and Gerber/drill export were not possible in this environment (no KiCad installation, no schematic in the repository).

1. Starting state vs. this routing pass

Item	Handoff says	Measured in repo file	Routed draft
Footprints	141	141	141
Pads	591	602	602
Signal nets	112	119 named nets with >= 2 pads	119
Tracks (segments)	0	0	12462
Vias	0	0	333
Copper zones (planes)	0	0	2 defined (GND In1.Cu, 3V3 In2.Cu), unfilled in file
Unconnected items	353	353 (KiCad, per handoff)	22

The handoff counts (591 pads, 112 nets) do not match the file (602 pads, 119 named nets). The file was used as the authority; the difference was not explained in the repository.

2. Checks run on the routed file (independent geometry check, shapely)

Check	Result	Notes
Shorts (different nets touching)	0	
Clearance < 0.15 mm (different nets, same layer)	0	router masks use 0.16 mm
Hole-to-hole < 0.25 mm	0	
Board-edge violations (< 0.30 mm)	0	
Dangling track ends	0	
Dangling vias	0	
Nets fully connected	107 / 119	GND / 3V3 counted connected when each SMD pad has a via into the plane
Unconnected pad items	22	section 4
Plane islands (GND In1 / 3V3 In2)	2 / 4	islands of the recomputed fill; see section 5
KiCad DRC / Gerbers / BOM / position file	not run	requires KiCad, which is not installed here

3. Layer usage

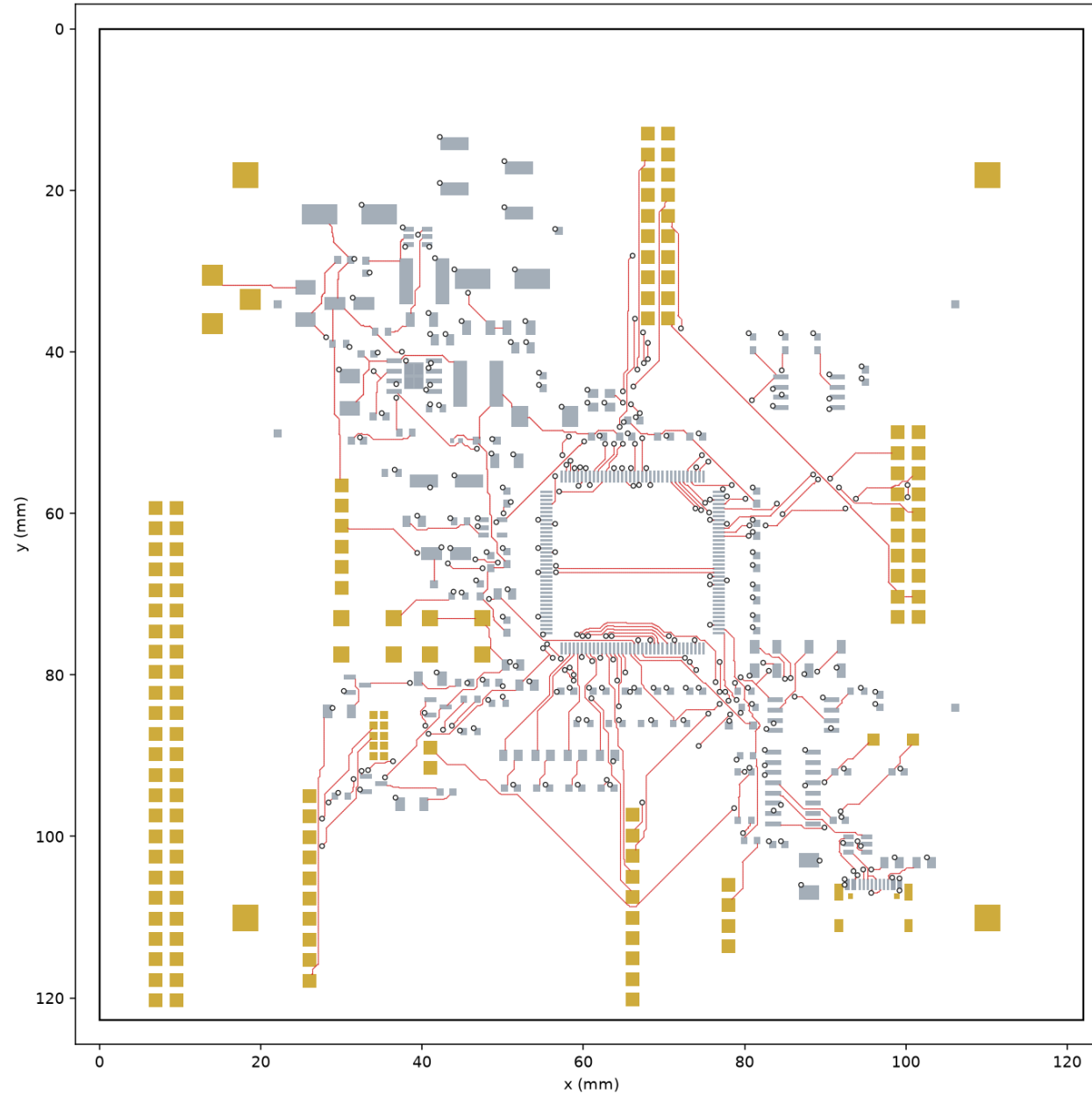
Layer	Role	Routed length (mm)	Via use
F.Cu	signals, FPGA fanout	1263.4	all 333 vias pass through every layer
In1.Cu	GND plane (zone, unfilled in file)	plane	reached by GND vias / through-hole pads

In2.Cu	3V3 plane (zone, unfilled in file)	plane	reached by 3V3 vias / through-hole pads
B.Cu	signals	1455.5	

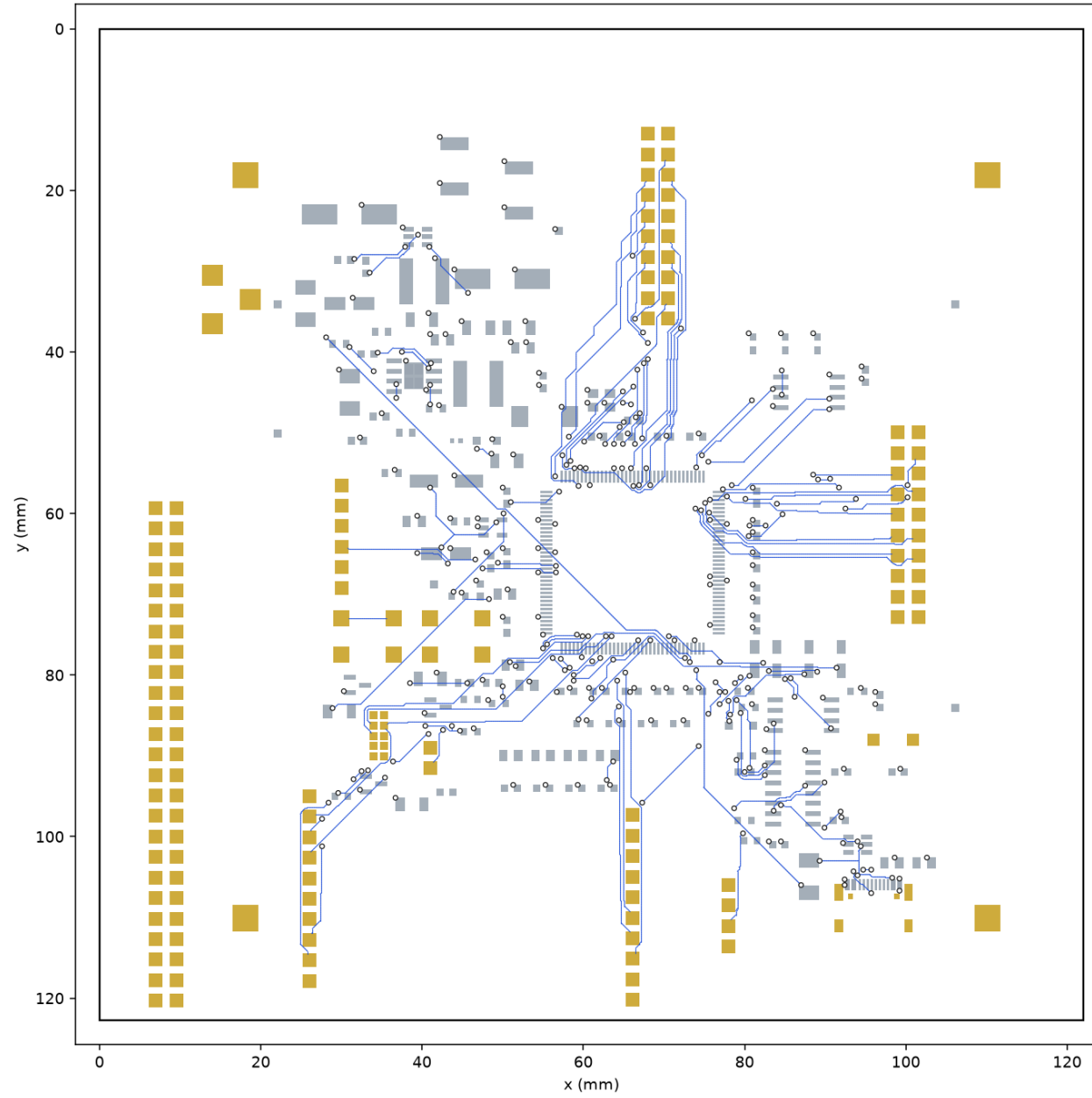
Track widths: signals 0.15 mm, power nets (VM_IN, VM_IN_RAW, J9_VIN, U5_SW, U6_SW, USB_VBUS, 5V_USB, 1V1, 2V5) 0.30 mm. Vias 0.50 mm / 0.25 mm drill. Clearance 0.15 mm. Copper-to-edge 0.30 mm.

Board figures (coordinates in mm, board origin at the top-left, as in KiCad)

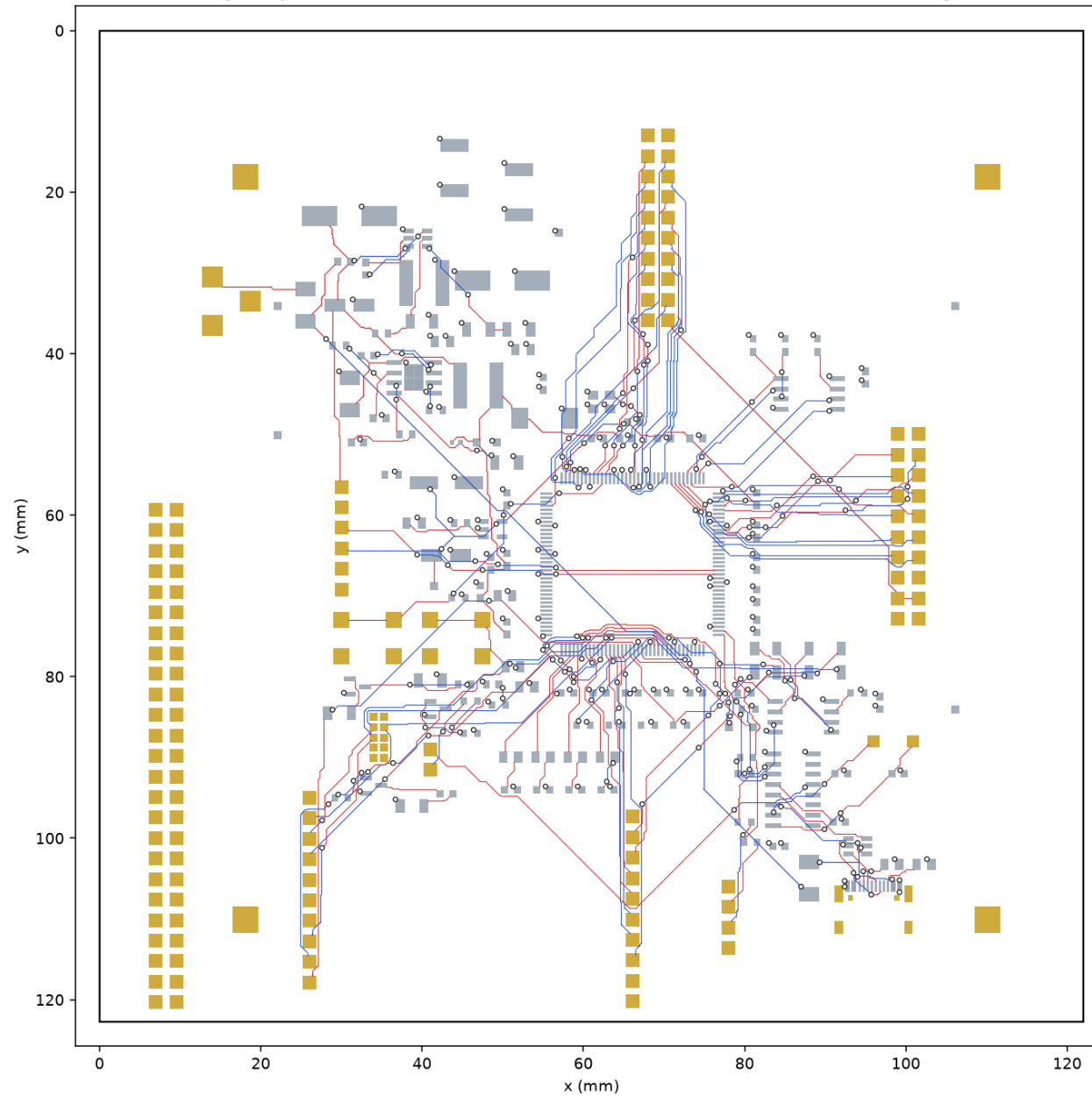
F.Cu (top) - routed copper; red = track, grey = SMD pad, gold = through-hole



B.Cu (bottom) - routed copper; blue = track



Both signal layers (red F.Cu, blue B.Cu), black dots = vias (In1 GND / In2 3V3 reached through them)



4. Open items (pad connections not routed)

Net	Pad(s) left unrouted	Note
1V1	U1.130, U1.20, U1.29, U1.38, U1.66, U1.83	FPGA core/IO supply pin, power-width track could not reach it
2V5	U1.17, U1.132, U1.96, U1.53	FPGA core/IO supply pin, power-width track could not reach it
A0	U1.45	
A1	U1.46	
B12	J4.15	
B13	J4.16	
B14	J4.17	
CFG_0	R6.1	
CFG_2	R7.1	
TMS	J2.2, J1.6	
motor_dir2	J6.5	
motor_fault_n	R19.1, J6.6	

The router runs three passes and keeps the best one. Raising the search budget from 350k to 1.5M expansions gave the same 22 failures, so these pins are blocked by copper already placed, not by search limits. The next step is rip-up-and-reroute of the blocking nets or a manual change to the FPGA escape order.

5. Unassigned FPGA pins (not routed - no net in the placement file)

48 of the 144 FPGA (U1, LFE5U-12F, LQFP-144) pads have no net in the placement file: 5, 6, 7, 10, 11, 12, 13, 14, 18, 19, 22, 23, 24, 25, 26, 27, 28, 30, 31, 33, 34, 35, 37, 67, 68, 69, 71, 72, 76, 77, 78, 79, 80, 81, 82, 84, 90, 91, 92, 93, 94, 95, 109, 118, 119, 120, 121, 144. They are most likely the FPGA's VCC, VCCIO and GND pins. The schematic that would confirm this is not in the repository, so no net was assigned and nothing was routed to them. The FPGA cannot be powered from this board until these pins are assigned from the schematic.

6. BOM / release items noted (not changed)

- L1 = 10 uH, Sunlord SWPA6045S (feeds U5, AP62300TWU-7 step-down).
- L2 = 10 uH, Sunlord SWPA6045S (feeds U6, MP1584EN). The handoff reports a cart record for 33 uH; the value was NOT changed. The MP1584 datasheet design table and the purchased part's current rating must be checked by a person.
- Y1 = 25 MHz active oscillator, Abracon ASE 4-pin 3.2x2.5 mm. Pin 1 / enable / output pin order must be checked against the purchased part; the footprint pinout was not verified against a datasheet here.
- U5/U6 passives, switching-node ringing, and inductor current ratings were not verified (no schematic or datasheet in the repository).

7. Every routed connection

Each row is one connection the router added. For signal nets the first pad of the net is the anchor; each row joins one more pad to the copper already connected to it. For GND and 3V3 each SMD pad gets its own via into the plane. 'Layers' is the copper used by that connection; 'Len' is the track length in mm; 'Vias' is the number of vias the connection uses.

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
1V1	0.30	C6.1 (tree)	C7.1	F.Cu	3.03	0	routed
1V1	0.30	C6.1 (tree)	C21.1	F.Cu	8.33	0	routed
1V1	0.30	C6.1 (tree)	L2.2	F.Cu	1.96	0	routed
1V1	0.30	C6.1 (tree)	TP3.1	F.Cu	4.44	0	routed
1V1	0.30	C6.1 (tree)	R38.1	F.Cu	2.67	0	routed
1V1	0.30	C6.1 (tree)	C2.1	F.Cu	15.87	0	routed
1V1	0.30	C6.1 (tree)	C3.1	F.Cu	1.20	0	routed
1V1	0.30	C6.1 (tree)	C4.1	F.Cu	2.91	0	routed
1V1	0.30	C6.1 (tree)	C5.1	F.Cu	4.74	0	routed
1V1	0.30	C6.1 (tree)	U1.38	-	-	-	OPEN
1V1	0.30	C6.1 (tree)	U1.66	-	-	-	OPEN
1V1	0.30	C6.1 (tree)	U1.130	-	-	-	OPEN
1V1	0.30	C6.1 (tree)	U1.83	-	-	-	OPEN
1V1	0.30	C6.1 (tree)	U1.29	-	-	-	OPEN
1V1	0.30	C6.1 (tree)	U1.20	-	-	-	OPEN
2V5	0.30	U7.5 (tree)	C42.1	B.Cu,F.Cu	8.37	2	routed
2V5	0.30	U7.5 (tree)	TP2.1	F.Cu	2.60	0	routed
2V5	0.30	U7.5 (tree)	C22.1	B.Cu,F.Cu	7.18	1	routed
2V5	0.30	U7.5 (tree)	C8.1	B.Cu,F.Cu	19.53	2	routed
2V5	0.30	U7.5 (tree)	C9.1	F.Cu	3.44	0	routed
2V5	0.30	U7.5 (tree)	C10.1	F.Cu	3.53	0	routed
2V5	0.30	U7.5 (tree)	C11.1	F.Cu	10.61	0	routed
2V5	0.30	U7.5 (tree)	U1.96	-	-	-	OPEN
2V5	0.30	U7.5 (tree)	U1.132	-	-	-	OPEN
2V5	0.30	U7.5 (tree)	U1.17	-	-	-	OPEN
2V5	0.30	U7.5 (tree)	U1.53	-	-	-	OPEN
5V_USB	0.30	J3.17 (tree)	J3.18	F.Cu	0.90	0	routed
5V_USB	0.30	J3.17 (tree)	J4.20	F.Cu	44.46	0	routed
A0	0.15	J5.1 (tree)	U1.45	-	-	-	OPEN
A1	0.15	J5.2 (tree)	R11.1	B.Cu,F.Cu	31.45	4	routed
A1	0.15	J5.2 (tree)	U1.46	-	-	-	OPEN

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
A10	0.15	U1.99 (tree)	J3.5	B.Cu,F.Cu	23.53	1	routed
A11	0.15	U1.104 (tree)	J3.6	B.Cu,F.Cu	25.14	2	routed
A12	0.15	U1.105 (tree)	J3.7	B.Cu,F.Cu	21.26	1	routed
A13	0.15	U1.106 (tree)	J3.8	B.Cu,F.Cu	26.58	1	routed
A14	0.15	U1.107 (tree)	J3.9	B.Cu,F.Cu	21.67	1	routed
A15	0.15	U1.108 (tree)	J3.10	B.Cu,F.Cu	25.44	2	routed
A2	0.15	R10.1 (tree)	U1.47	B.Cu,F.Cu	40.88	8	routed
A2	0.15	R10.1 (tree)	J5.3	B.Cu,F.Cu	27.89	3	routed
A3	0.15	J5.4 (tree)	U1.48	F.Cu	28.00	0	routed
A4	0.15	J5.5 (tree)	U1.49	B.Cu,F.Cu	32.57	2	routed
A4	0.15	J5.5 (tree)	R12.1	B.Cu,F.Cu	30.62	1	routed
A5	0.15	J5.6 (tree)	U1.50	B.Cu,F.Cu	32.69	1	routed
A6	0.15	J5.7 (tree)	U1.51	B.Cu,F.Cu	35.64	1	routed
A7	0.15	J5.8 (tree)	U1.52	B.Cu,F.Cu	38.24	1	routed
A8	0.15	U1.97 (tree)	J3.3	B.Cu,F.Cu	25.07	4	routed
A9	0.15	U1.98 (tree)	J3.4	B.Cu,F.Cu	29.70	5	routed
B0	0.15	U1.135 (tree)	J4.3	B.Cu,F.Cu	40.80	2	routed
B1	0.15	U1.136 (tree)	J4.4	B.Cu,F.Cu	42.40	3	routed
B10	0.15	U1.126 (tree)	J4.13	B.Cu,F.Cu	27.56	3	routed
B11	0.15	U1.127 (tree)	J4.14	B.Cu,F.Cu	28.43	1	routed
B12	0.15	U1.1 (tree)	J4.15	-	-	-	OPEN
B13	0.15	U1.2 (tree)	J4.16	-	-	-	OPEN
B14	0.15	U1.3 (tree)	J4.17	-	-	-	OPEN
B15	0.15	U1.4 (tree)	J4.18	B.Cu,F.Cu	30.64	1	routed
B2	0.15	U1.139 (tree)	J4.5	B.Cu,F.Cu	46.07	1	routed
B3	0.15	U1.140 (tree)	J4.6	B.Cu,F.Cu	51.49	1	routed
B4	0.15	U1.141 (tree)	J4.7	B.Cu,F.Cu	39.03	3	routed
B5	0.15	U1.142 (tree)	J4.8	B.Cu,F.Cu	38.23	2	routed
B6	0.15	U1.143 (tree)	J4.9	B.Cu,F.Cu	34.82	3	routed
B7	0.15	U1.128 (tree)	J4.10	B.Cu,F.Cu	37.80	2	routed
B8	0.15	U1.124 (tree)	J4.11	B.Cu,F.Cu	30.48	3	routed
B9	0.15	U1.125 (tree)	J4.12	B.Cu,F.Cu	33.09	1	routed
CC1	0.15	R36.1 (tree)	J8.A5	B.Cu,F.Cu	3.48	2	routed
CC2	0.15	R37.1 (tree)	J8.B5	F.Cu	3.91	0	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
CCLK	0.15	U1.54 (tree)	R8.1	F.Cu	10.35	0	routed
CCLK	0.15	U1.54 (tree)	R9.1	B.Cu,F.Cu	11.52	2	routed
CFG_0	0.15	U1.62 (tree)	R6.1	-	-	-	OPEN
CFG_1	0.15	U1.59 (tree)	R5.1	B.Cu,F.Cu	13.99	2	routed
CFG_2	0.15	U1.58 (tree)	R7.1	-	-	-	OPEN
D1_A	0.15	D1.2 (tree)	R33.2	F.Cu	2.95	0	routed
D6_K	0.15	D6.1 (tree)	R34.1	F.Cu	4.51	0	routed
D7_A	0.15	D7.2 (tree)	R44.2	B.Cu,F.Cu	1.80	2	routed
DONE	0.15	U1.56 (tree)	TP5.1	B.Cu,F.Cu	50.32	10	routed
DONE	0.15	U1.56 (tree)	R4.1	F.Cu	4.67	0	routed
DONE	0.15	U1.56 (tree)	R32.1	B.Cu,F.Cu	27.69	4	routed
DONE	0.15	U1.56 (tree)	J1.9	B.Cu,F.Cu	22.38	1	routed
DTR_F	0.15	JP1.2 (tree)	R25.1	B.Cu,F.Cu	8.93	1	routed
FB_1V1	0.15	R38.2 (tree)	R39.1	B.Cu,F.Cu	4.24	2	routed
FB_1V1	0.15	R38.2 (tree)	U6.4	F.Cu	10.28	0	routed
INITN	0.15	U1.55 (tree)	TP6.1	F.Cu	17.88	0	routed
INITN	0.15	U1.55 (tree)	R3.1	F.Cu	18.88	0	routed
INITN	0.15	U1.55 (tree)	R31.1	B.Cu,F.Cu	25.69	2	routed
INITN	0.15	U1.55 (tree)	J1.10	F.Cu	32.42	0	routed
J10_RX	0.15	J10.3 (tree)	R22.2	B.Cu,F.Cu	12.85	1	routed
J10_TX	0.15	J10.2 (tree)	R24.2	F.Cu	8.18	0	routed
J9_VIN	0.30	J9.1 (tree)	D8.2	F.Cu	9.22	0	routed
PROGRAMN	0.15	U1.57 (tree)	R2.1	B.Cu,F.Cu	30.18	2	routed
PROGRAMN	0.15	U1.57 (tree)	SW2.1	F.Cu	5.03	0	routed
PROGRAMN	0.15	U1.57 (tree)	SW2.1	F.Cu	4.70	0	routed
PROGRAMN	0.15	U1.57 (tree)	Q1.3	F.Cu	9.15	0	routed
PROGRAMN	0.15	U1.57 (tree)	J1.4	B.Cu,F.Cu	24.37	1	routed
Q1_G	0.15	Q1.1 (tree)	R26.1	B.Cu,F.Cu	8.15	2	routed
Q1_G	0.15	Q1.1 (tree)	R25.2	B.Cu,F.Cu	4.66	1	routed
Q2_B	0.15	Q2.1 (tree)	R32.2	B.Cu,F.Cu	3.64	2	routed
Q2_C	0.15	Q2.3 (tree)	D1.1	F.Cu	6.37	0	routed
Q4_B	0.15	Q4.1 (tree)	R31.2	F.Cu	3.53	0	routed
Q4_C	0.15	Q4.3 (tree)	D6.2	B.Cu,F.Cu	12.10	2	routed
TCK	0.15	U1.63 (tree)	J2.5	B.Cu,F.Cu	45.72	1	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
TCK	0.15	U1.63 (tree)	J1.8	B.Cu,F.Cu	27.17	1	routed
TDI	0.15	U1.61 (tree)	J2.3	B.Cu,F.Cu	42.81	1	routed
TDI	0.15	U1.61 (tree)	J1.3	B.Cu,F.Cu	15.75	1	routed
TDO	0.15	U1.60 (tree)	J2.4	B.Cu,F.Cu	38.34	1	routed
TDO	0.15	U1.60 (tree)	J1.2	B.Cu	16.05	0	routed
TMS	0.15	U1.64 (tree)	J2.2	-	-	-	OPEN
TMS	0.15	U1.64 (tree)	J1.6	-	-	-	OPEN
U2_CLK	0.15	U2.6 (tree)	R9.2	B.Cu,F.Cu	13.30	2	routed
U2_CS	0.15	R13.1 (tree)	U2.1	B.Cu,F.Cu	6.13	2	routed
U2_CS	0.15	R13.1 (tree)	R12.2	B.Cu,F.Cu	12.02	2	routed
U2_DI	0.15	R10.2 (tree)	U2.5	B.Cu,F.Cu	11.87	2	routed
U2_DO	0.15	R11.2 (tree)	U2.2	F.Cu	7.98	0	routed
U2_HOLD	0.15	R15.1 (tree)	U2.7	F.Cu	2.28	0	routed
U2_WP	0.15	R14.1 (tree)	U2.3	B.Cu,F.Cu	17.63	2	routed
U3_HOLD	0.15	U3.7 (tree)	R18.1	F.Cu	4.44	0	routed
U3_WP	0.15	R17.1 (tree)	U3.3	B.Cu,F.Cu	5.20	2	routed
U4_DTR	0.15	JP1.1 (tree)	U4.13	B.Cu,F.Cu	59.80	2	routed
U4_RXD	0.15	R23.1 (tree)	U4.3	B.Cu,F.Cu	5.98	2	routed
U4_TXD	0.15	R21.1 (tree)	U4.2	B.Cu,F.Cu	3.47	2	routed
U4_V3	0.15	C34.1 (tree)	U4.4	F.Cu	7.96	0	routed
U5_BST	0.15	U5.6 (tree)	C38.1	B.Cu,F.Cu	9.11	2	routed
U5_EN	0.15	U5.5 (tree)	R49.1	F.Cu	9.78	0	routed
U5_EN	0.15	U5.5 (tree)	R48.2	F.Cu	2.09	0	routed
U5_FB	0.15	R47.1 (tree)	R46.2	F.Cu	1.00	0	routed
U5_FB	0.15	R47.1 (tree)	U5.4	B.Cu,F.Cu	12.24	2	routed
U5_SW	0.30	U5.2 (tree)	L1.1	F.Cu	2.62	0	routed
U5_SW	0.30	U5.2 (tree)	C38.2	F.Cu	3.80	0	routed
U6_BST	0.15	U6.8 (tree)	C33.1	B.Cu,F.Cu	7.84	2	routed
U6_COMP	0.15	R45.1 (tree)	U6.3	B.Cu,F.Cu	5.70	2	routed
U6_COMP_RC	0.15	R45.2 (tree)	C35.1	F.Cu	7.56	0	routed
U6_EN	0.15	U6.2 (tree)	C31.1	F.Cu	5.54	0	routed
U6_EN	0.15	U6.2 (tree)	R41.2	B.Cu,F.Cu	5.66	2	routed
U6_FREQ	0.15	U6.6 (tree)	R40.1	B.Cu,F.Cu	2.70	2	routed
U6_SW	0.30	L2.1 (tree)	U6.1	F.Cu	7.00	0	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
U6_SW	0.30	L2.1 (tree)	C33.2	F.Cu	2.91	0	routed
U6_SW	0.30	L2.1 (tree)	D11.1	F.Cu	5.76	0	routed
U7_EN	0.15	U7.3 (tree)	R43.2	F.Cu	1.56	0	routed
U7_EN	0.15	U7.3 (tree)	C32.1	F.Cu	7.24	0	routed
USB_DN	0.15	U8.3 (tree)	J8.B7	F.Cu	3.80	0	routed
USB_DN	0.15	U8.3 (tree)	J8.A7	F.Cu	1.22	0	routed
USB_DN_F	0.15	U8.4 (tree)	U4.6	B.Cu,F.Cu	13.03	2	routed
USB_DP	0.15	U8.1 (tree)	J8.A6	B.Cu,F.Cu	9.30	2	routed
USB_DP	0.15	U8.1 (tree)	J8.B6	F.Cu	1.32	0	routed
USB_DP_F	0.15	U8.6 (tree)	U4.5	F.Cu	12.13	0	routed
USB_VBUS	0.30	U8.5 (tree)	J8.A4	B.Cu,F.Cu	4.49	2	routed
USB_VBUS	0.30	U8.5 (tree)	J8.B9	F.Cu	0.00	0	routed
USB_VBUS	0.30	U8.5 (tree)	J8.A9	B.Cu,F.Cu	4.57	1	routed
USB_VBUS	0.30	U8.5 (tree)	J8.B4	F.Cu	0.00	0	routed
USB_VBUS	0.30	U8.5 (tree)	D9.2	B.Cu,F.Cu	4.90	1	routed
VM_IN	0.30	U5.3 (tree)	FB1.2	B.Cu,F.Cu	7.12	2	routed
VM_IN	0.30	U5.3 (tree)	C39.1	F.Cu	4.84	0	routed
VM_IN	0.30	U5.3 (tree)	R41.1	F.Cu	3.90	0	routed
VM_IN	0.30	U5.3 (tree)	R48.1	F.Cu	4.39	0	routed
VM_IN	0.30	U5.3 (tree)	C23.1	F.Cu	4.95	0	routed
VM_IN	0.30	U5.3 (tree)	U6.7	B.Cu,F.Cu	8.14	2	routed
VM_IN	0.30	U5.3 (tree)	J6.1	F.Cu	16.59	0	routed
VM_IN_RAW	0.30	FB1.1 (tree)	D8.1	F.Cu	7.40	0	routed
VM_IN_RAW	0.30	FB1.1 (tree)	D9.1	B.Cu,F.Cu	97.60	4	routed
Y2_XI	0.15	C36.1 (tree)	Y2.1	F.Cu	5.13	0	routed
Y2_XI	0.15	C36.1 (tree)	U4.7	B.Cu,F.Cu	9.42	2	routed
Y2_XO	0.15	Y2.2 (tree)	C37.1	F.Cu	3.77	0	routed
Y2_XO	0.15	Y2.2 (tree)	U4.8	B.Cu,F.Cu	16.30	2	routed
clk_25m	0.15	U1.133 (tree)	Y1.3	B.Cu,F.Cu	12.81	4	routed
ext_rst_n	0.15	U1.134 (tree)	C1.1	B.Cu,F.Cu	12.49	2	routed
ext_rst_n	0.15	U1.134 (tree)	R1.1	B.Cu,F.Cu	13.17	2	routed
ext_rst_n	0.15	U1.134 (tree)	SW1.1	F.Cu	6.69	0	routed
ext_rst_n	0.15	U1.134 (tree)	SW1.1	B.Cu	4.70	0	routed
flash_cs_n	0.15	R16.1 (tree)	U3.1	F.Cu	3.68	0	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
flash_cs_n	0.15	R16.1 (tree)	U1.111	B.Cu,F.Cu	15.48	2	routed
flash_miso	0.15	U1.113 (tree)	U3.2	B.Cu,F.Cu	14.94	2	routed
flash_mosi	0.15	U1.112 (tree)	U3.5	B.Cu,F.Cu	21.32	2	routed
flash_sck	0.15	U1.110 (tree)	U3.6	B.Cu,F.Cu	19.95	2	routed
led0	0.15	U1.39 (tree)	R27.1	F.Cu	15.21	0	routed
led0_k	0.15	R27.2 (tree)	D2.1	F.Cu	3.33	0	routed
led1	0.15	U1.40 (tree)	R28.1	F.Cu	13.76	0	routed
led1_k	0.15	D3.1 (tree)	R28.2	F.Cu	3.33	0	routed
led2	0.15	U1.41 (tree)	R29.1	F.Cu	12.52	0	routed
led2_k	0.15	D4.1 (tree)	R29.2	F.Cu	3.33	0	routed
led3	0.15	U1.44 (tree)	R30.1	F.Cu	13.67	0	routed
led3_k	0.15	R30.2 (tree)	D5.1	B.Cu,F.Cu	3.46	2	routed
motor_dir1	0.15	U1.89 (tree)	J6.4	B.Cu,F.Cu	46.34	3	routed
motor_dir2	0.15	U1.102 (tree)	J6.5	-	-	-	OPEN
motor_fault_n	0.15	U1.103 (tree)	R19.1	-	-	-	OPEN
motor_fault_n	0.15	U1.103 (tree)	J6.6	-	-	-	OPEN
pwm_out	0.15	U1.88 (tree)	J6.3	B.Cu,F.Cu	47.58	4	routed
spi0_cs_n	0.15	U1.115 (tree)	J3.11	B.Cu,F.Cu	29.51	1	routed
spi0_miso	0.15	U1.117 (tree)	J3.14	B.Cu,F.Cu	35.44	1	routed
spi0_mosi	0.15	U1.116 (tree)	J3.13	B.Cu,F.Cu	31.42	1	routed
spi0_sck	0.15	U1.114 (tree)	J3.12	B.Cu,F.Cu	33.23	1	routed
uart_rx	0.15	U1.73 (tree)	R21.2	B.Cu,F.Cu	18.03	2	routed
uart_rx	0.15	U1.73 (tree)	R22.1	F.Cu	5.74	0	routed
uart_tx	0.15	U1.74 (tree)	R23.2	F.Cu	17.35	0	routed
uart_tx	0.15	U1.74 (tree)	R24.1	F.Cu	10.36	0	routed
3V3	0.15	In2.Cu plane	R17.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C15.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C41.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	U7.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	Q4.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	TP1.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	J5.10	pad-via	0.00	0	routed
3V3	0.15	In2.Cu plane	R16.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R13.2	F.Cu	0.10	1	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
3V3	0.15	In2.Cu plane	R46.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	J10.1	pad-via	0.00	0	routed
3V3	0.15	In2.Cu plane	U1.9	F.Cu	0.40	1	routed
3V3	0.15	In2.Cu plane	U1.16	F.Cu	0.40	1	routed
3V3	0.15	In2.Cu plane	U1.36	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	U1.43	F.Cu	1.19	1	routed
3V3	0.15	In2.Cu plane	U1.70	F.Cu	0.40	1	routed
3V3	0.15	In2.Cu plane	U1.86	F.Cu	0.40	1	routed
3V3	0.15	In2.Cu plane	U1.100	F.Cu	0.40	1	routed
3V3	0.15	In2.Cu plane	U1.122	F.Cu	0.40	1	routed
3V3	0.15	In2.Cu plane	U1.137	F.Cu	0.40	1	routed
3V3	0.15	In2.Cu plane	J2.1	pad-via	0.00	0	routed
3V3	0.15	In2.Cu plane	C18.1	F.Cu	0.20	1	routed
3V3	0.15	In2.Cu plane	U3.8	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C12.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C17.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	J1.1	pad-via	0.00	0	routed
3V3	0.15	In2.Cu plane	R14.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	J3.1	pad-via	0.00	0	routed
3V3	0.15	In2.Cu plane	R8.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C14.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	D2.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C24.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C40.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C25.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C20.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	D3.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C28.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R18.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R2.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	L1.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R4.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	D4.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R15.2	F.Cu	0.10	1	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
3V3	0.15	In2.Cu plane	U4.16	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R3.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	D5.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R5.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R43.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	U2.8	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C29.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R19.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C26.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	Y1.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	Y1.4	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R33.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C30.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R1.2	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C16.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C27.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	R44.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C19.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	C13.1	F.Cu	0.10	1	routed
3V3	0.15	In2.Cu plane	J4.1	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	R47.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C15.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C41.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	U8.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	U7.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R36.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	TP4.1	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C6.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C34.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J5.9	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	Q2.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	U5.1	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C5.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J10.4	pad-via	0.00	0	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
GND	0.15	In1.Cu plane	C36.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	U1.8	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	U1.15	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	U1.21	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	U1.32	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	U1.42	F.Cu	0.40	1	routed
GND	0.15	In1.Cu plane	U1.65	F.Cu	0.40	1	routed
GND	0.15	In1.Cu plane	U1.75	F.Cu	0.40	1	routed
GND	0.15	In1.Cu plane	U1.85	F.Cu	0.40	1	routed
GND	0.15	In1.Cu plane	U1.87	F.Cu	0.40	1	routed
GND	0.15	In1.Cu plane	U1.101	F.Cu	0.40	1	routed
GND	0.15	In1.Cu plane	U1.123	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	U1.129	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	U1.131	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	U1.138	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	J2.6	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J2.10	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	C18.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	D11.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	U3.4	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C12.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C2.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C1.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C17.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C39.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C21.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C7.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J1.7	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J3.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J3.15	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J3.16	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J3.19	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J3.20	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	C11.2	F.Cu	0.10	1	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
GND	0.15	In1.Cu plane	C35.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R34.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R6.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	Q1.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	SW1.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	SW1.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	R49.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C14.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J9.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	U6.5	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	U6.9	F.Cu	0.30	1	routed
GND	0.15	In1.Cu plane	C24.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C42.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R37.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C40.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C25.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C20.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R26.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C28.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J8.A1	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J8.A12	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J8.B1	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J8.B12	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J8.SH	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J8.SH	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J8.SH	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J8.SH	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	D7.1	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C9.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C3.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	U4.1	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C23.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C10.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C4.2	F.Cu	0.10	1	routed

Net	Width	From (anchor/plane)	To pad	Layers	Len mm	Vias	Status
GND	0.15	In1.Cu plane	U2.4	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C32.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C29.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C37.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R40.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C26.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	Y1.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J6.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	C30.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C16.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R39.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	SW2.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	SW2.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	C27.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C19.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C13.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C8.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	J4.2	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	J4.19	pad-via	0.00	0	routed
GND	0.15	In1.Cu plane	C31.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	R7.2	F.Cu	0.10	1	routed
GND	0.15	In1.Cu plane	C22.2	F.Cu	0.10	1	routed

8. Not done / limits of this draft

- No KiCad installation here: DRC, zone refill and project-open check were not run. Open the file in KiCad 10 and press B (fill all zones), then run DRC.
- The two plane zones are defined but not filled in the file; the fill I recomputed for checking is not written to the board.
- Thermal reliefs for plane connections are not drawn in my model; KiCad creates them when the zones are filled.
- The router uses a 0.10 mm grid with 45-degree steps; track geometry was verified with exact shape clearance, not with KiCad's engine.
- Rotated-footprint pads use KiCad's rotation convention ($x' = x \cos + y \sin$, $y' = -x \sin + y \cos$), confirmed against KiCad's source discussion; a KiCad check is still required.
- No schematic was available: the 112/119 net list is taken from the PCB pad net names, not compared to a schematic.
- Courtyard routing rule was not enforced. Tracks may pass over component bodies where pad clearance allows.